Deep brain stimulation of symptom-specific networks in Parkinson's disease

Paper: Rajamani et al. (2024). Multicentre connectomic DBS study.

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Page references below use PDF page numbers, including front matter.

To which general question does this paper relate?

What DBS can control and how to choose the performance objective. Examines whether different symptoms benefit from stimulation of different networks.

How does it answer the general questions?

Evidence. Stimulation locations and tract engagement were related to changes in tremor, rigidity, bradykinesia and axial symptoms. The study included 232 retrospective patients and five additional prospective patients (PDF pp. 1–10).

Usable insight. Symptom improvement maps to partly distinct networks. The proposed network-blending approach weights these networks according to an individual's symptom profile when choosing stimulation settings. This supports a symptom-dependent performance objective rather than assuming one physiological feature captures all treatment benefit (PDF pp. 7–10).

Model relevance. Tract engagement can inform hypotheses about where control enters a network and which outputs matter. It does not directly determine the notebook's temporal gains, delays, nonlinearities, or measurement matrix. Connectomic associations and the small prospective demonstration do not establish a universal stimulation-to-symptom transfer function; normative tractography is a stated limitation.

What papers does it cite that could provide more information?

- Horn et al. (2017), *Connectivity Predicts Deep Brain Stimulation Outcome in Parkinson Disease*: connectivity-based outcome prediction (ref. 19, PDF p. 14).
- Roediger et al. (2023), *Automated deep brain stimulation programming based on electrode location*: randomized crossover programming trial (ref. 41, PDF p. 15).